

CZT Detector Modeling for Coded Aperture X-ray Diffraction Imaging Applications

J. Greenberg, K. Iniewski, and D. Brady

Abstract—CZT detectors are primary candidates for many next-generation X-ray diffraction imaging systems. We have conducted a study to evaluate performance tradeoffs and limitations of CZT energy-sensitive detectors in coded aperture X-ray imaging (CAXI) architectures. We have found, as expected, that performance of the coded aperture architecture depends critically on energy resolution (ER) of the detector. We have also found that spectrum tail, as measured for mono-energetic sources, is less important although the incomplete charge collection leads to distorted system level information. We have built a mathematical model for the energy spectrum that can be used to optimize the detector geometry (pixel size, detector thickness), detector fabrication (injecting vs. non-injecting contacts) and operating conditions (HV bias, count rate) for the optimum performance of the coded aperture X-ray imaging systems.

I. INTRODUCTION

X-ray spectroscopy has long been used for non-destructive examination of unknown objects. While traditional transmission-based X-ray imaging systems record only object densities, this information is typically insufficient to perform material discrimination. Nevertheless, a variety of applications in medical, security, and industrial imaging require or can benefit from the measurement of the material composition of an object at each location within its volume. To realize this spatially-resolved material discrimination, a number of approaches have been developed, including fluorescence, hyper-spectral transmission, back scattering, phase contrast, and coherent scatter imaging. From these techniques, coherent scatter imaging has been the most promising approach because of its capacity to provide material-specific signatures in optically thick materials. In coherent scatter, X-rays scatter elastically from the electron cloud of an atom or molecule. The interference of X-rays scattered from different locations in the material yields sensitivity to the microscopic structure of the material and

produces a unique signature for a variety of materials. All previous studies have assumed ideal detector response. We study here the influence of non-ideal effects of the detector's response on a CAXI system consisting of an incident X-ray pencil beam and a linear array of energy-sensitive pixels. CZT charge-sharing between pixels, incomplete charge collection due to DOI and K-edge escape have been taken into account. This work leverages previous research at Duke University on computational X-ray imaging through the CAXI program [1,2] and Redlen's previous work on analytical modeling of CZT for Nuclear Medicine applications[3-5].

II. CAXI MODELING METHODOLOGY

To study the impact of a non-ideal detector response, we conduct system level numerical simulations of a pencil beam CAXI system in which the scattered x-rays pass through a coded aperture and are detected by a linear array of CZT detector pixels. This geometry is described and the analytical model developed in [2]. In these simulations, we assume an incident, broadband X-ray spectrum produced by a conventional tungsten anode X-ray tube operating at 125 kVp. This enables us to calculate the distribution of scattered X-rays in a mean sense (i.e., we calculate the mean number of photons at each energy scattered into a particular solid angle).

To translate this into a realistic measurement, we implement a model of the detector and detection process. For example, we take into account the finite size of the pixel (here we choose 0.8 mm square pixels) and an energy response described by a mean probability distribution (this is a first-order model, which is the most straightforward to implement). As an example, a photon arriving at 60keV produces a Gaussian response at 60keV with a constant tail of some amplitude (defined as the fractional tail amplitude). 70keV photon produces a similar response at 70keV that adds to that response generated by the 60keV photon. We then include Poisson noise by taking this mean value at each energy and use the Poisson probability density function to realize a particular photon count at each energy based on Poisson statistics and the mean calculated value.

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J. Greenberg is with Duke University, Durham, NC, 27708 USA (e-mail: jag27@duke.edu).

K. Iniewski is with Redlen Technologies, Saanich, BC V8M 1X6S, Canada (e-mail: kris.iniewski@redlen.com).

D. Brady is with Duke University, Durham, NC, 27708 USA (e-mail: dbrady@duke.edu).

III. EFFECT OF ENERGY RESOLUTION

We first consider the effect of detector energy resolution first. Figure 1 shows raw data for a point object for three different values of FWHM (2.5keV, 4keV, 7keV) with the same energy bin size (2.3keV). Figure 2 shows the mean (and standard deviation) of the normalized mean squared error (m.s.e.) obtained by performing maximum likelihood estimation (MLE) for 50 realizations of random objects. The system response is dependent on the FWHM value as expected; namely, the normalized mean squared error (m.s.e.) decreases with a decreasing FWHM energy response (e.g. reducing the FWHM from 7keV down to 2.5 keV improves m.s.e. by a factor of 2). The result that better energy resolution yields better signal conditioning (even with no improvement in sampling) can also be seen through in Fig. 3, which shows that the singular value spectrum is larger and flatter for detectors with better energy resolution indicating superior conditioning for inversion even in the presence of noise.

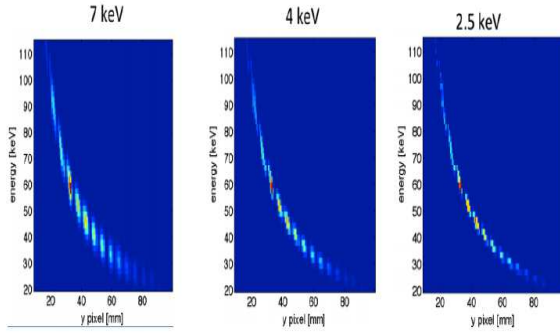


Fig.1 CAXI raw dataset for a point object for three cases of FWHM of 7keV, 4keV and 2.5keV.

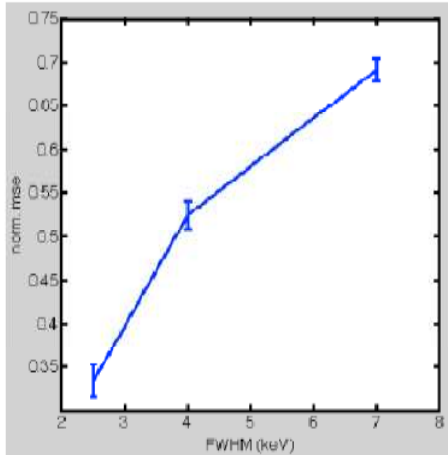


Fig.2 Simulated CAXI m.s.e. error dependence on FWHM of the detector. Error bars correspond to one standard deviation variation among the objects and Poisson realizations.

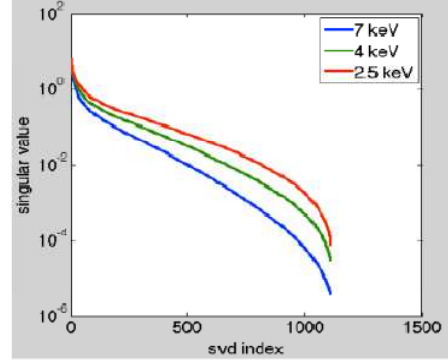


Fig.3 Simulated CAXI SVD for three different values of FWHM.

IV. EFFECT OF ENERGY TAILING

We look also at the effect of spectrum tailing (i.e. the inclusion of a non-Gaussian low-energy tail extending beyond the photo-peak). To evaluate different physically realizable cases, we consider different levels of spectrum tailing of 1%, 10%, 20% and 50% (defined as the fraction of the photo-peak amplitude). For all of these scenarios, we fix the FWHM value at 7keV and keep the total number of counts constant. It is possible to achieve tailing of 1% using charge-sharing correction or discrimination schemes; however, the maximum count rate of the system is in this case limited. A much higher level of tailing is typically observed if charge-shared events are not corrected, which is the case typically. For larger pixel sizes the amount of tailing can be as low as 10% whereas, for smaller pixels, the tail amplitudes can be as large as 50%. In general, the exact numbers depend on the pixel size, detector thickness, applied high voltage bias (HV), electron-hole mobility products of the bulk material, and photon count rate, and can be calculated using our analytical CZT model.

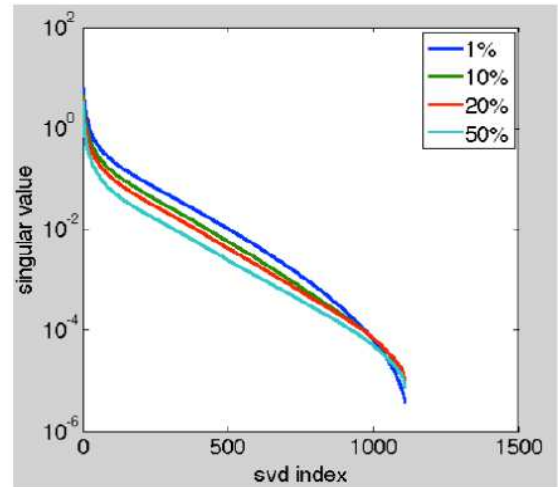


Fig.4 Simulated CAXI SVD for four different values of detector tailing.

As shown in Fig.4, the amplitudes of the singular values indicate slightly better performance when the tailing is reduced. However, improvements in the overall m.s.e value are rather small, as shown in Fig.5. For example, complete elimination of the 20% tail (by charge sharing correction) improves m.s.e error from 0.78 down to 0.69.

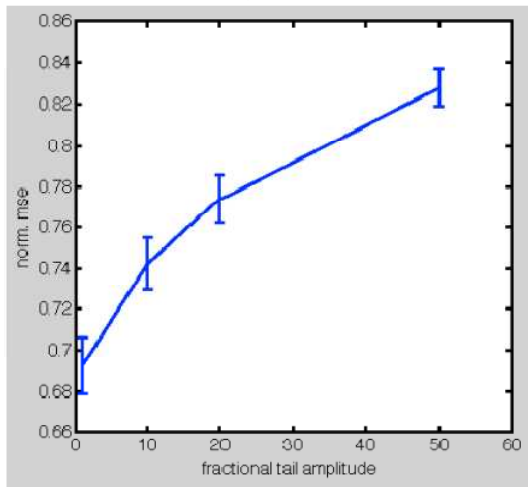


Fig. 5 Simulated CAXI rms error dependence on the amount of tailing. Error bars correspond to one standard deviation variation among the objects and Poisson realizations.

Our results indicate that the amount of spectrum tailing does not affect CAXI ability to reconstruct the object to large extent. The results were somewhat unexpected as prior to this study we assumed that detector tail non-idealities would have a major effect on the response. Part of the explanation comes from the fact that, with only 10 or 20% of the peak pulse amplitude, the total count difference in the peak is small (it depends on the peak energy, but for 40-80 keV this typically only amounts to a factor of 2-3 in peak amplitude reduction). As the signal is distributed over many pixels/energy bins, this small decrease in the statistics in one bin does not too much affect the overall ability to perform estimation when the statistics of the measurement process are accurately known. It is however possible that different metrics beyond m.s.e. over the whole object (i.e., material and location) such as resolution, might be affected more. For example, the presence of the tails impacts the estimation of the coherent scatter form factor without adversely affecting the spatial resolution, which may indicate why the fractional performance degradation is not so large with tails. Thus, while the tails do not strongly degrade the resulting tomographic image, the ability to classify the material at each location is reduced. Furthermore, it is expected that the role of tails will be more pronounced in the case of highly-multiplexed signals in the Poisson-limited regime [6] and when strong self-attenuation is present.

V. CONCLUSIONS

The energy resolution (ER) of the detector makes a big difference in the performance of the studied CAXI architecture. Improving the FWHM detector energy response from 7keV down to 2.5keV reduces the m.s.e. error by a factor of 2 and improves the momentum transfer resolution. As the energy resolution tends to degrade at high count rates, future work is required to optimize the count rate vs. ER trade-off. In addition, we find that charge sharing, if left uncorrected, leads to lower detector efficiency, increased spectrum tail, and decreased energy resolution. However, our system level simulation results indicate that charge sharing and other physical effects that cause spectrum tailing contribute only weakly to the performance of the CAXI architecture selected in the study. These results imply a high level of CAXI robustness with respect to detector non-ideal effects and bode very well for practical implementations of CAXI systems in material identification applications. In light of our current understanding of the device physics and signal processing, some recommendations for the future X-ray diffraction systems are being developed.

REFERENCES

- [1] J. Greenberg, K. Krishnamurthy, and D. Brady, *Snapshot Molecular Imaging using Coded Energy-Sensitive Detection*, vol. 21, no. 21, Optics Express, 2013.
- [2] J. Greenberg, K. Krishnamurthy, and D. Brady, *Compressive Single-Pixel Snapshot X-Ray Diffraction Imaging*, Optics Letters, vol. 39, pp. 111-114, 2014.
- [3] K. Iniewski, H. Chen, G. Bindley, I. Kuvvetli, C. Budtz-Jorgensen, *Modeling Charge-Sharing Effects in Pixellated CZT Detectors*, IEEE NSS-MICS, 2007.
- [4] K. Iniewski, S. Awadallah, F. Harris, H. Chen, G. Bindley, and P. Luke, *High Voltage Optimization in CZT Detectors*, IEEE NSS-MICS, 2008.
- [5] K. Iniewski, C. Seifert, F. Harris, S. Awadallah, H. Chen, and G. Bindley, *CZT Pixel Scaling for Improved Spatial Resolution in Medical Imaging*, IEEE NSS-MICS, 2008.
- [6] J.A. Greenberg, M. Hassan, K. Krishnamurthy, and D. Brady, *Structured illumination for tomographic X-ray diffraction imaging*, 139, 709-713, 2014.